Project: California Housing Market Intelligence

Overview

In this **six-week team** project, you will work with a real-world geospatial dataset to predict median house values across California. The project follows the complete data science pipeline, including team coordination, critical data investigation, regression modeling, error mitigation, and a final professional presentation.
Each week builds on the previous one. Therefore, **strong collaboration, documentation, and version control practices** established in Week 1 are essential for success.

Project Context

Your team is acting as real estate data analysts for a state-wide housing authority. You are provided with the 1990 California Census dataset (Download HERE), which contains complex spatial relationships and market anomalies.

Your goal is not only to build a high-performing regression model but also to provide **Explainable AI (xAI)** insights into why certain regions command premium prices. You must move beyond "black-box" accuracy to identify the specific economic and geographic drivers that definitively influence housing valuations.

Technical Objectives

Foundational Workflow

- **Data Investigation:** Analyze spatial features (ocean proximity) and economic correlations to identify predictive signals.
- **Baseline Experiment:** Construct a "Green-to-Gold" regression pipeline, including stratified partitioning and standard normalization.
- **Error Diagnosis:** Identify systematic overprediction in high-value properties (the $500k cap).

Advanced Pipeline Development

- **Feature Engineering:** Handle skewed distributions and multi-collinearity to improve model signal.
- **Model Optimization:** Transition from simple Linear Regression to Random Forest models to reduce Mean Absolute Error (MAE).
- **Binary Classification:** Deploy logistic regression to predict categorical geographic traits (e.g., "Is the house near the ocean?") based on economic features.

Weekly Project Roadmap

Week	Phase	Deliverable	Objective
1	**Infrastructure**	Group Collaboration Document	Establish communication, GitHub repository, and team roles.
2	**Profiling**	Data Audit & EDA	Explore dataset skewness, missing values, and spatial outliers.
3	**Investigation**	5-Question Analytical Report	Use KNIME to solve 5 critical market mysteries and justify variables.
4	**Modeling**	End-to-End Prediction Pipeline	Build a baseline regression model and diagnose overprediction bias.
5	**Correction**	Fixed & Optimized Pipeline	Fix skewness and implement Random Forest to lower MAE below $35k.
6	**Presentation**	Final Presentation Slides	Present methodology, model v1 vs. v2 comparison, and insights.

Key Requirements

- **Fully Automated Workflow:** Your team must build a fully automated KNIME pipeline that runs from raw `housing.csv` to final predictions without manual intervention.
- **Team Weekly Syn**: Each week, please fill out the form and use the form to reflect your weekly group meeting(s).
- **Economic Justification:** Your final report must identify which variables are redundant based on linear correlation values rather than intuition.

Week-by-Week Deliverables

Week 1 — Team Formation & Infrastructure

Establish roles, GitHub access, and communication norms. See the **Deliverable 1 instructions** for full group agreement requirements.

Week 2 — Data Understanding & Profiling

Deliverable 2: Data Audit Dashboard & Team Discussion Form. Identify the "shape" of the housing data. Perform a technical distribution audit (Histograms/Box Plots) and identify the impact of the house value "cap" at $500,001.

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Week 3 — Critical Thinking & Data Investigation

Deliverable 3: 5-Question Analytical Report. Solve the following mysteries using labelled Metanodes and Annotations:

1. **Ocean Proximity:** Is it a unique driver or a proxy for income? (Box Plots/Scatter Plots).

   Nodes: GroupBy, Box Plot, Scatter Plot, Annotation. Form your hypothesis in an Annotation before running anything, argue your conclusion in a second Annotation after.

2. **Redundancy:** Which variable would you remove from the model based on **Linear Correlation**?

   Nodes: Linear Correlation, Scatter Matrix, Column Filter, Annotation. Pick one variable, justify it with the correlation values — not intuition.

3. **Outliers:** Pinpoint a neighborhood that is a geographic outlier and speculate on real-world causes. Nodes: Scatter Plot, Row Filter, Statistics, Annotation. Speculate on real-world causes in the Annotation.

4. **Regional Logic:** Is the income-to-value relationship identical in Northern and Southern California?

   Nodes: Row Splitter, Linear Correlation, Scatter Plot, Line Plot, Annotation. Split on latitude > 37.5 for North/South.

5. **Market Caps:** At what income level does the house value "flatline," and is this a data artifact?

   Nodes: Scatter Plot, Row Filter, GroupBy, Histogram, Annotation. Explain whether the cap is a real market phenomenon or a data artefact.

Week 4 — Pipeline Creation & Model Investigation

Deliverable 4: End-to-End KNIME Prediction Pipeline. **Tasks:**

- **Pipeline Construction:** Build a workflow:

  CSV Reader → Missing Value → Row Filter → Math Formula → One to Many → Normalizer → Partitioning → Linear Regression Learner → Regression Predictor → Numeric Scorer

- **Bias Check:** Your model overpredicts houses above $400k. Use **GroupBy** and **Box Plots** to explain why.

  Nodes: Math Formula, GroupBy, Box Plot, Annotation.

- **Target Swap:** Build a sub-model to predict `median_income` instead of house value to identify feature importance.Nodes: Column Filter, Linear Regression Learner, Regression Predictor, Numeric Scorer, Bar Chart, Annotation.

Week 5 — Fix the Model & Optimization

Deliverable 5: Corrected Pipeline. **Tasks:**

- ***Address overprediction above $400k:* Nodes: Histogram, Row Filter, Math Formula, Numeric Scorer, Tile View, Annotation.**
- **Skew Mitigation:** Use **Math Formula** (Log transforms) to address skewed histograms found in Week 2.
- **Error Reduction:** Implement a **Random Forest Learner**. Your goal is to reduce **MAE below $35,000**. Nodes: Numeric Scorer, Random Forest Learner, Annotation.
- **Classification Task:** Use **Logistic Regression** to predict if a house is "NEAR OCEAN." Compare results with Random Forest.

  Nodes: Rule Engine, Column Filter, One to Many, Logistic Regression Learner, Scorer, Random Forest Learner, Tile View, Annotation.

Week 6 — Final Presentation

Deliverable 6: Final Presentation Slides & Final Workflow. Present a professional summary including:

- Comparison of Linear Regression (v1) vs. Random Forest (v2).
- Logic behind log transformations and skewness fixes.
- Demonstration of the final KNIME pipeline
- Limitations and future improvements (including honest reflection on the 5 homes the model was "most wrong" about or remaining errors)
- Contributions of each team member